Similar questions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A right circular cone C of height 3 m and base radius 4 m has its base fixed to a horizontal plane. One end of a light elastic string of natural length 1.5 m and modulus 50 N is fixed to the vertex of C. The other end is attached to a particle P of mass 3 kg. P moves in a horizontal circle with constant speed in contact with the smooth curved surface of C. The extension of the string is 1 m.

(a) Find the tension in the string. **[2]**

(b) Find the speed of P. **[7]**

2 One end of a light elastic string of natural length 0.8 m and modulus λ N is attached to a fixed point O on a smooth horizontal table. The other end is attached to a particle A of mass 0.5 kg which moves on the table in a circle with constant speed $v\,\mathrm{m\,s^{-1}}$. The extension of the string is x m.

(a) Show that $\lambda x^2 + 0.8\lambda x - 0.4v^2 = 0$. **[3]**

(b) For $\lambda = 120$ and $v = 4$, find x. **[2]**

(c) Show that for large λ, $\lambda x \approx mv^2$ and explain what this means for an inextensible string. **[3]**

3 A trainee of mass 80 kg stands on the edge of a platform. A light inextensible cord of length 4 m joins the trainee to a hook H at the same level as the platform and 4 m horizontally away, with the cord straight and taut. The trainee is projected downwards with speed $5\,\mathrm{m\,s^{-1}}$ perpendicular to the cord and swings in a vertical circle. The ground is 6 m below the level of H. Model the trainee as a particle and ignore air resistance.

(a) The cord is released from H when the speed of the trainee is $8\,\mathrm{m\,s^{-1}}$. Show that the tension just before release is 1670 N (3 s.f.). **[6]**

(b) Find the speed and the angle to the horizontal at which the trainee reaches the ground. **[7]**

Question 1

(a) $T = \dfrac{50 \times 1}{1.5} = 33.3$ N

(b) P is 2.5 m from the vertex along the slope; semi-vertical angle α has $\sin\alpha = \frac{4}{5}$, $\cos\alpha = \frac{3}{5}$, so the radius is $r = 2.5 \times 0.8 = 2$ m. Vertically: $T\cos\alpha + R\sin\alpha = 3g$: $20 + 0.8R = 29.4$, $R = 11.75$ N. Horizontally (inwards): $T\sin\alpha - R\cos\alpha = \dfrac{3v^2}{2}$: $26.67 - 7.05 = 1.5v^2$, $v = 3.62\ \mathrm{m\,s^{-1}}$

Question 2

(a) $T = \dfrac{\lambda x}{0.8}$ provides the centripetal force at radius $0.8 + x$: $\dfrac{\lambda x}{0.8} = \dfrac{0.5v^2}{0.8+x}$, so $\lambda x(0.8+x) = 0.4v^2$

(b) $120x^2 + 96x - 6.4 = 0$, $x = 0.0619$ m

(c) $\lambda x^2 + \lambda l x = lmv^2$ gives $x = \frac{l}{2}\left(-1 + \sqrt{1 + \frac{4mv^2}{\lambda l}}\right) \approx \frac{l}{2} \cdot \frac{2mv^2}{\lambda l} = \frac{mv^2}{\lambda}$, so $\lambda x \approx mv^2$, i.e. the tension is mv^2/l (the radius is effectively l): exactly the tension in an inextensible string of length l

Question 3

(a) Height fallen $d = \dfrac{8^2 - 5^2}{2g} = 1.99$ m, so the cord makes angle ϕ below the horizontal with $\sin\phi = \frac{1.99}{4} = 0.497$.

Radially towards H: $T - mg\sin\phi = \dfrac{mv^2}{4}$, $T = 80\,(16 + 9.8(0.497)) = 1670$ N

(b) Velocity is perpendicular to the cord: $v_x = 8\sin\phi = 3.98$, downward $8\cos\phi = 6.94$. Remaining fall $6 - 1.99 = 4.01$ m: $4.01 = 6.94t + 4.9t^2$, $t = 0.441$ s. Final downward speed $6.94 + 9.8(0.441) = 11.26$; speed $= \sqrt{3.98^2 + 11.26^2} = 11.9\ \mathrm{m\,s^{-1}}$ at 70.5° below the horizontal